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Proteomic dataset of sorghum leaf and root responses to single and combined drought and heat stress

Ali Elnaeim Elbasheir Ali¹, Robert E. Sharp², Laura Greeley³, Scott C. Peck³, David L. Tabb⁴ & Ndiko Ludidi^{1,5,6}

Drought and heat stress significantly limit crop growth and productivity. Their simultaneous occurrence, as often observed in summer crops, leads to larger yield losses. Sorghum is well adapted to dry and hot conditions. Despite the progress that has been made in determining proteomic responses to water deficit or heat stress in crops, such information remains limited for crops subjected to combined water deficit and heat stress. This study presents quantitative proteomics analyses of leaf and root tissues of two contrasting sorghum genotypes grown under normal conditions, water deficit stress, heat stress, and water deficit combined with heat stress. We identified differentially expressed proteins between the two genotypes under these different treatments. GO and KEGG annotation revealed biological processes and molecular pathways associated with sorghum responses to these treatments. Interpretation as well as integration of these proteomics data with other 'omics' signatures may highlight key mechanisms involved in sorghum adaptations to these stresses.

Background & Summary

Drought and heat stress are major challenges to crop production and food security worldwide. These stresses affect crucial aspects of plant growth and development such as flower production and grain filling, and consequently lead to substantial yield reduction^{1–3}. High temperature and drought can occur simultaneously in the field, which can severely impact plant growth and development. Various studies demonstrated that single and combined drought and heat may result in similar effects on plants. For example, reduction in photosynthesis and transpiration rate were reported in crops exposed to both single and combined drought and heat stress⁴. However, the response of plants to the combined effects of different environmental stresses is not necessarily additive; instead, a unique response may emerge due to the synergistic or antagonistic effects of both stressors. For example, unique protective responses were observed in *Eucalyptus globulus* under combined heat and drought, which were not activated in response to the single stresses⁵. Plant adaptation to combined stresses are governed by the interplay between the two stressors and their respective signalling pathways, where one pathway may inhibit the other, or one pathway may be prioritized over the other⁵. For example, the decrease in stomatal conductance and rate of transpiration due to water deficit may induce heat stress as the leaf temperature rises, while the increase in transpiration because of heat stress may lead to significant leaf water loss. Consequently, when the plant experiences these two stresses simultaneously, a balance must be maintained between minimizing water loss and dissipating heat. This interconnected relationship highlights the complexity of plant responses to combined stresses. In addition, the interaction of genetic, biochemical and physiological factors contributes to responses to single stresses. This interaction further complicates plant responses to the combined effects of different stresses⁶.

Sorghum (*Sorghum bicolor* [L.] Moench subsp. *bicolor*), known for its tolerance to drought and heat stress, thrives in arid and semi-arid tropics. Sorghum can be productive in low-input, hot, and dry conditions where other major grain crops such as maize, wheat and rice may struggle under such abiotic stresses⁷. Sorghum resilience to these abiotic stresses, together with the availability of its genome sequence, can help to screen for, and

¹Department of Biotechnology, University of the Western Cape, Bellville, South Africa. ²Division of Plant Science and Technology, University of Missouri, Columbia, MO, USA. ³Division of Biochemistry, University of Missouri, Columbia, MO, USA. ⁴Department of Structural Biology and Chemistry, Institut Pasteur, Paris, France. ⁵Faculty of Agriculture and Natural Sciences, University of Mpumalanga, Mbombela, South Africa. ⁶DSI-NRF Centre of Excellence in Food Security, University of the Western Cape, Bellville, South Africa. ✉e-mail: Ndiko.Ludidi@ump.ac.za

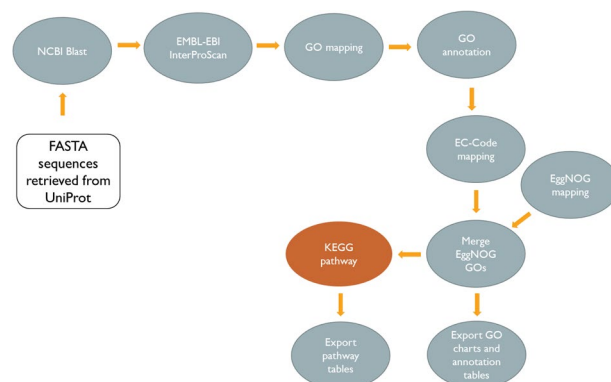


Fig. 1 Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses pipeline using the Blast2GO tool implemented in OmicsBox software version 3.0.30 (BioBam Bioinformatics¹³).

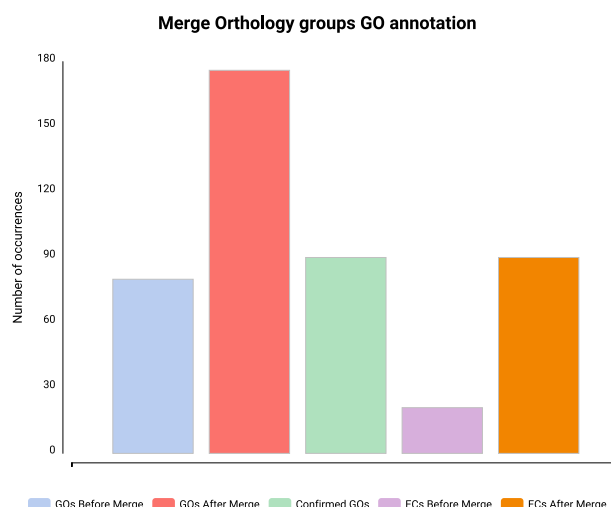


Fig. 2 The new Gene Ontology (GO) terms and Enzyme Commission (EC) codes, generated from the EggNOG tool within OmicsBox software version 3.0.30 (BioBam Bioinformatics¹³), were added to the existing annotations obtained through the Blast2GO program implemented in the same software version.

functionally annotate, genes that govern molecular mechanisms associated with drought and heat stress tolerance^{8,9}. Understanding these mechanisms is particularly significant to produce economically vital crop varieties that face threats posed by these two stresses⁷. Proteomic analyses offer a systematic and comparative approach to unveiling protein expression profiles across various plant tissues and physiological states. To our knowledge, there is no published study investigating proteomic responses in contrasting sorghum genotypes exposed to combined drought and heat stress, especially at the reproductive stage. Leaves play distinct roles from roots, and the leaf proteome, which may differ from the root proteome, may also influence plant response to these stresses. Therefore, it is critical to study both organs to gain a comprehensive understanding of stress responses in plants¹⁰. In this study, we used label-free quantification (LFQ) with data-dependant acquisition (DDA) to compare changes in the protein expression profiles of the leaves and roots of two contrasting sorghum genotypes grown under semi-field conditions (in tunnels erected on a field plot). The two genotypes were subjected to water deficit, heat stress, or combined water deficit and heat stress. The data generated from this research have the potential to uncover candidate proteins that modulate sorghum molecular adaptation mechanisms to these stresses. Such insights can facilitate the development of sorghum varieties and potentially other crops with enhanced tolerance to single and combined water deficit and heat stress.

Methods

Plant materials and growth conditions. Two contrasting sorghum genotypes, namely Alisto (tolerant genotype) and ICSB (sensitive genotype), were used in this study. The selection of these genotypes was based on a previous experiment carried out in our laboratory, which screened multiple sorghum genotypes for their tolerance to single and combined drought and heat stress. The current study was conducted at the agricultural research station of the Plant Stress Tolerance Laboratory (PSTL) in Lukholweni, located in the Matatiele Local Municipality, Alfred Nzo District Municipality, Eastern Cape, South Africa. The geographic coordinates of the research station are approximately -30.631891, 28.858739 (i.e. 30°37'54.8"S 28°51'31.5"E). The experiment was

Time (min)	Flow rate ($\mu\text{L}/\text{min}$)	Solvent A (%)	Solvent B (%)
0.00	0.30	98.0	2.00
0.00	0.30	98.0	2.00
3.00	0.30	98.0	2.00
8.00	0.30	95.0	5.00
48.0	0.30	82.0	18.0
58.0	0.30	70.0	30.0
60.0	0.30	20.0	80.0
70.0	0.30	20.0	80.0
70.1	0.30	98.0	2.00
85.0	0.30	98.0	2.00

Table 1. Peptide separation gradient setup.

Full Scan	
Resolution	70,000 (@ m/z 200)
AGC target value	3e6
Scan range	350–2000 m/z
Maximal injection time (ms)	100
Data-dependent MS/MS	
Inclusion	Off
Resolution	17,500 (@ m/z 200)
AGC target value	1e5
Maximal injection time (ms)	50
Loop Count	10
Isolation window width (m/z)	3
NCE (%)	27
Data-dependent Settings	
Underfill ratio (%)	1
Charge exclusion	Unassigned, 1, 7, 8, > 8
Peptide match	preferred
Exclusion isotopes	on
Dynamic exclusion (s)	60

Table 2. Details of mass spectrometry data acquisition parameters.

performed using a completely randomized block design. Planting was carried out during the 2020–2021 cropping season, starting on November 15, using a rain shelter structure (tunnel) with high polycarbonate roof design and polycarbonate walls to effectively prevent water entry while ensuring good aeration.

Seeds of the two sorghum genotypes were surface sterilized in 0.35% sodium hypochlorite (NaOCl) for 10 minutes, then rinsed five times with distilled water prior to sowing into the soil. Each genotype was planted in a three-row plot, 2 meters in length, with a row spacing of 20 cm. Seeds were sown at a depth of 5 cm, maintaining a 20-cm distance between seeds within the rows. The experiment was performed using two rain shelters (tunnels), one for heat treatments and the other for non-heat treatments. All plants were irrigated with tap water at field capacity until reaching the five-leaf stage (five leaves that are fully expanded with visible collars). At this stage (Stage 2), plants were subjected to either the control condition (no stress applied) or to the water deficit condition. Water deficit stress was induced by withholding water until the day of harvesting. The heat treatment was applied when symptoms of drought stress, namely leaf rolling, were observed in both genotypes. For the sensitive genotype, this occurred 50 days after withholding water, while the tolerant genotype took 70 days to exhibit leaf rolling. Plants that were designated for heat and the combined treatment (water deficit and heat) were exposed to a temperature ranging from 38 to 42 °C for 6 hours daily (from 10:00 to 16:00) over a two-week period. The heat was applied using an LPG-driven conventional heating system in a fan-assisted tunnel. For the control and water deficit plants, the air temperature was maintained in a range of 25 to 30 °C. The difference in water deficit stress duration between the two sorghum genotypes reflects their differing drought tolerance capacities. By allowing each genotype to experience a similar degree of stress, the study captures the distinct responses of each genotype under stress. Plants were harvested immediately after the completion of the heat treatment, with most plants entering the hard dough stage (Stage 8), although a minority of them were still at the soft dough stage (Stage 7). The water status of the plants was determined by measuring the relative water content (RWC) on the youngest fully expanded leaves of the plants according to the following formula:

Recorded protein search configuration	
Rule	Value
Protein database	Sorghum_UP000000768_413 80prot_201021.fasta
Spectrum-level FDR	Auto cut
Cleavage residues	RK
Digest cutter	C-terminal cutter
Peptide termini	Fully specific
Maximum number of missed cleavages	2
Fragmentation type	CID low energy
Precursor tolerance	10.0 ppm
Fragment tolerance	20.0 ppm
Charges applied to charge-unassigned spectra:	1, 2, 3
Precursor mass max	10000.0
N-glycan search	None
O-glycan search	None
Skip bad spectrum	true
Off by x isotopes	-2, -1, 0, +1, +2
Contaminants added	false
Decoys added	true
Additional parameters:	
Disulfide Enable	false
Trisulfide Enable	false
DSS Crosslink Enable	false
Custom Crosslink Enable	false
Wildcard Enable	false
Combyne cut off score	Auto
Protein FDR cutoff	1%
Focused DB created	false
Export mzIdentML	false
Score version	2
precursor_assignment_flags	2
po_NumberMonosReturn	2
Modification searches:	
common_modifications_max	1
rare_modifications_max	1
Fixed and variable modifications:	
Oxidation/+15.994915 @ M	common2
Deamidated/+0.984016 @ N, Q	common2
Carbamidomethyl/+57.021464 @ C	fixed
Product Version	PMI-Byonic-Com:v3.8.13

Table 3. Details of search engine parameters.

$$RWC(\%) = 100 \times \frac{[FW - DW]}{[TW - DW]};$$

Where FW is the fresh weight, TW is the turgid weight and DW is the dry weight. The growth parameters recorded were shoot length and stem diameter. Harvested tissues (four youngest leaves, including the basal growth zone and the mature blade, and the whole root system excavated from a soil depth of 30 cm) were rapidly washed in distilled water to remove soil debris prior to grinding into a powder in liquid nitrogen. The harvested material was stored at -80 °C for further use. Chlorophyll content was measured spectrophotometrically on leaf material of the two genotypes as described by Tait and Hik¹¹.

Protein quantification with Label-Free LC-MS/MS. Protein extracts from leaf and root tissues were obtained from five biological replicates of each sorghum genotype under each treatment using a modified SDS/phenol method as described by Ali *et al.*¹². To create each biological replicate, tissues from five randomly selected plants were pooled together.

Protein solubilisation, quantification, on-bead digestion, and label-free LC-MS/MS analyses were performed as detailed by Ali *et al.*¹². Protein pellets were resuspended in 50 mM triethylammonium bicarbonate (TEAB)/2%

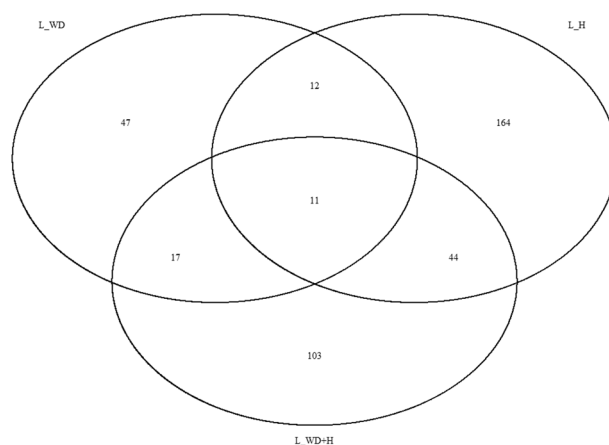


Fig. 3 The numbers of differentially expressed proteins between the two sorghum genotypes (Alisto and ICSB) in the leaves (L) in response to water deficit (WD), heat stress (H) and combined water deficit and heat stress (WD + H).

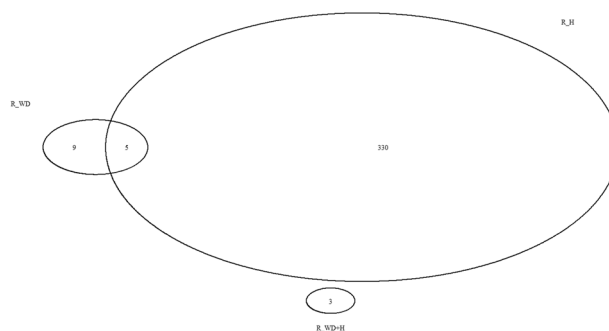


Fig. 4 The numbers of differentially expressed proteins between the two sorghum genotypes (Alisto and ICSB) in the roots (R) in response to water deficit (WD), heat stress (H) and combined water deficit and heat stress (WD + H).

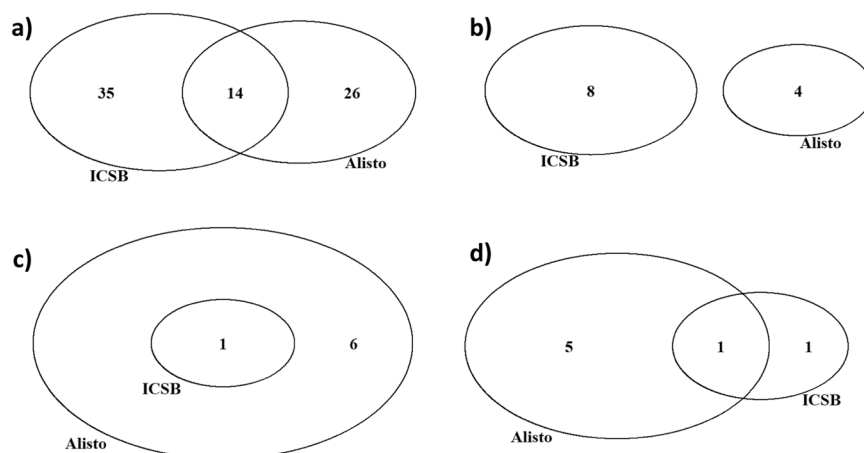


Fig. 5 Venn diagram of the numbers of uniquely upregulated and downregulated proteins in the two sorghum genotypes (Alisto and ICSB) in response to water deficit stress (WD). (a) Proteins with increased expression in the leaves; (b) Proteins with decreased expression in the leaves; (c) Proteins with increased expression in the roots; (d) Proteins with decreased expression in the roots.

SDS buffer by incubating at 95 °C for 5 min, followed by centrifugation for 5 min at 10,000 × g. Proteins quantification was performed using the QuantiPro BCA assay kit (Sigma) according to the manufacturer's protocol.

In preparation for hydrophilic interaction liquid chromatography (HILIC), magnetic beads were washed twice with 250 µL of washing buffer [15% acetonitrile (ACN) and ammonium acetate (100 mM), pH 4.5] for

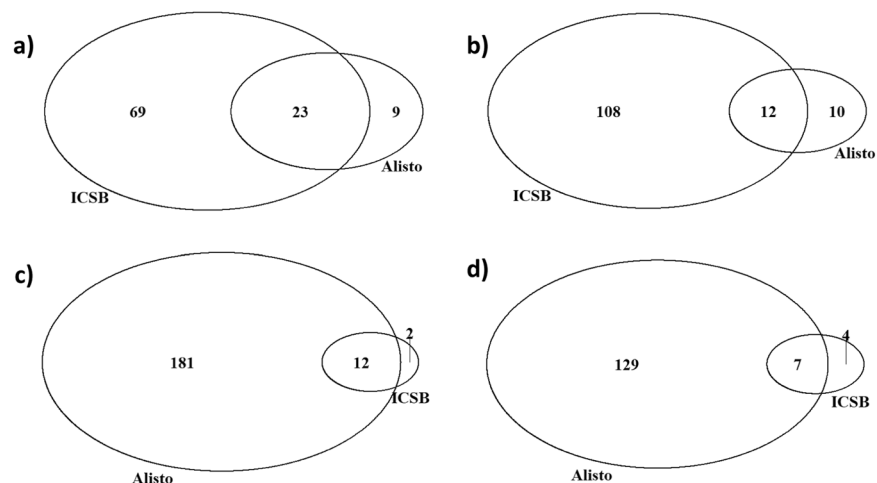


Fig. 6 Venn diagram of the numbers of uniquely upregulated and downregulated proteins in the two sorghum genotypes (Alisto and ICSB) in response to heat stress (H). (a) Proteins with increased expression in the leaves; (b) Proteins with decreased expression in the leaves; (c) Proteins with increased expression in the roots; (d) Proteins with decreased expression in the roots.

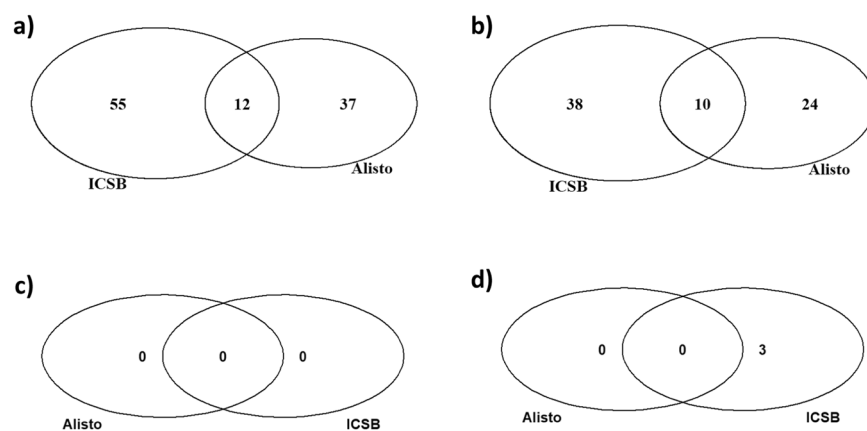


Fig. 7 Venn diagram of the numbers of uniquely upregulated and downregulated proteins in the two sorghum genotypes (Alisto and ICSB) in response to combined water deficit and heat stress (WD + H). (a) Proteins with increased expression in the leaves; (b) Proteins with decreased expression in the leaves; (c) Proteins with increased expression in the roots; (d) Proteins with decreased expression in the roots.

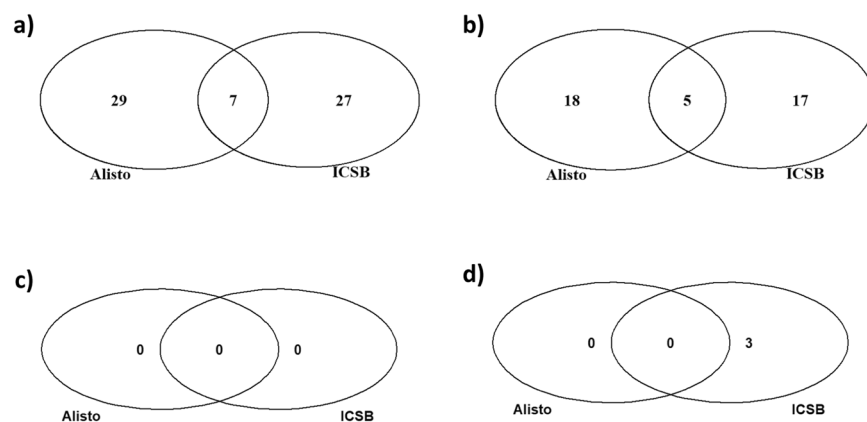


Fig. 8 Venn diagram of the numbers of uniquely upregulated and downregulated proteins in the two sorghum genotypes (Alisto and ICSB) that are only occurring under combined water deficit and heat stress (WD + H). (a) Proteins with increased expression in the leaves; (b) Proteins with decreased expression in the leaves; (c) Proteins with increased expression in the roots; (d) Proteins with decreased expression in the roots.



Fig. 9 Presentation of differentially expressed proteins between two contrasting sorghum genotypes under water deficit stress as observed in the leaves. Each column represents the log₂-fold change of differentially expressed proteins in each genotype. Expression differences are illustrated in different colors. Red color indicates upregulation, black color indicates no change in expression and green color indicates downregulation.

1 min each time. The beads were then resuspended in a loading buffer containing 30% ACN and ammonium acetate (200 mM) at pH 4.5. Proteins were reduced with 10 mM Tris (2-carboxyethyl) phosphine (TCEP) for 1 h at 60 °C. Protein samples were alkylated with 10 mM methyl methanethiosulphonate (MMTS) for 15 min at room temperature and subsequently digested with trypsin (Promega, Madison, WI, USA) at a 1:10 ratio (trypsin:protein) at 37 °C for 4 h. The resulting peptides were desalted and resuspended in loading buffer (2.5% ACN, 0.1% formic acid (FA)).

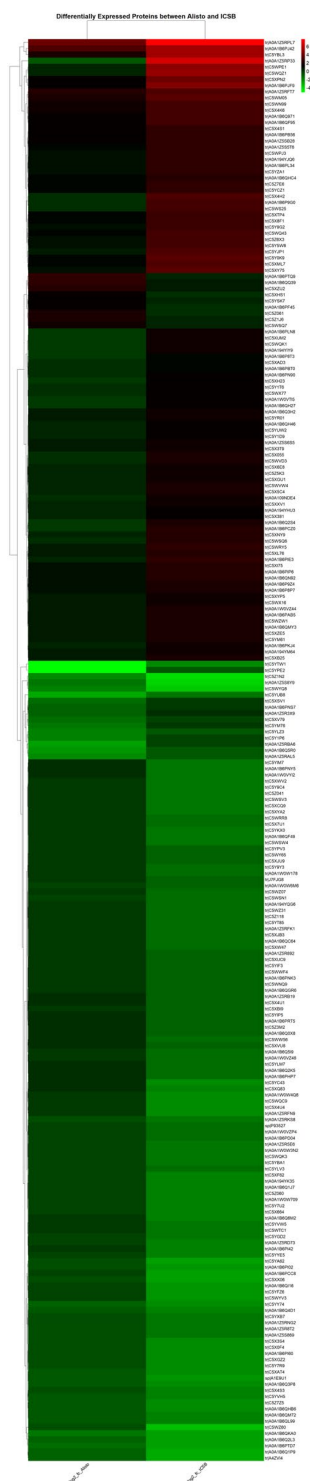


Fig. 10 Presentation of differentially expressed proteins between two contrasting sorghum genotypes under heat stress as observed in the leaves. Each column represents the log₂-fold change of differentially expressed proteins in each genotype. Expression differences are illustrated in different colors. Red color indicates upregulation, black color indicates no change in expression and green color indicates downregulation.

LC-MS/MS analyses was conducted on a Q-Exactive quadrupole Orbitrap mass spectrometer (Thermo Fisher Scientific, Waltham, MA, USA) coupled with a nano-HPLC system (Dionex Ultimate 3000). The desalted digests were dissolved in 2% ACN, 0.1% formic acid (Sigma) and loaded onto a C18 trap column (PepAcclaim) in 3.5% solvent B (0.1% FA, 0.1% ACN) at a flow rate of 5 μ L/min for 4 min. The solvent system used was solvent A [0.1% FA in liquid chromatography (LC) water (Burdick and Jackson)] and solvent B (0.1% FA with ACN). The multi-step gradient for peptide separation was generated at 300 nL/min as follows: 2–5% Solvent B over 5 min,

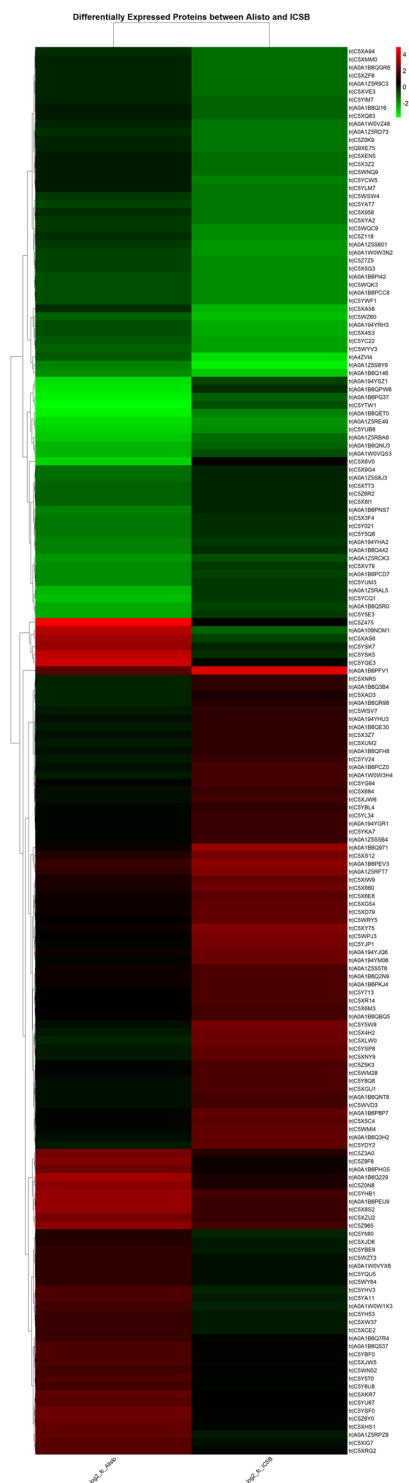


Fig. 11 Presentation of differentially expressed proteins between two contrasting sorghum genotypes under combined water deficit and heat stress as observed in the leaves. Each column represents the log₂-fold change of differentially expressed proteins in each genotype. Expression differences are illustrated in different colors. Red color indicates upregulation, black color indicates no change in expression and green color indicates downregulation.

5–18% Solvent B over 40 min, 18–30% Solvent B over 10 min, 30–80% Solvent B over 2 min. The gradient was then finished with 80% Solvent B for 10 min, followed by 15 mins at 2% Solvent B. The mass spectrometer was performed with a capillary temperature set at 320 °C on a positive ion mode at a +1.95 kV electrospray.

For peptide and protein identification, database search was carried out using Byonic software v3.8.13 (Protein Metrics, USA). MS/MS spectra were searched against a sorghum reference proteome

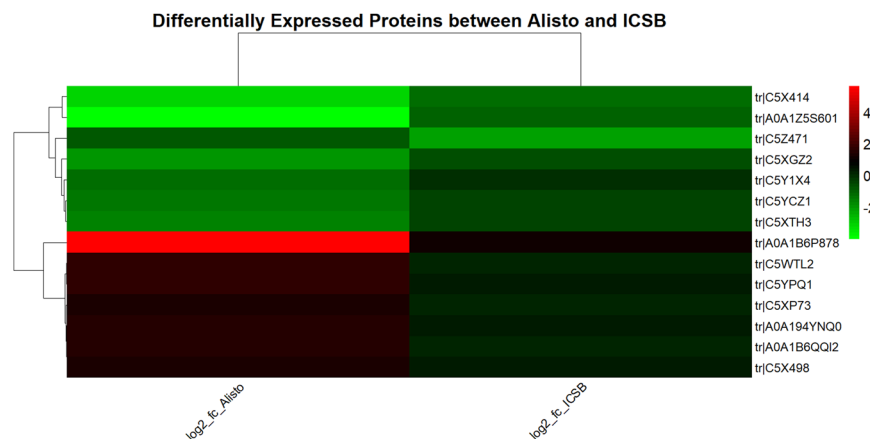


Fig. 12 Presentation of differentially expressed proteins between two contrasting sorghum genotypes under water deficit stress as observed in the roots. Each column represents the log2-fold change of differentially expressed proteins in each genotype. Expression differences are illustrated in different colors. Red color indicates upregulation, black color indicates no change in expression and green color indicates downregulation.

(Sorghum_UP000000768_41380prot_201021.fasta) sourced from Uniprot (www.uniprot.org) on October 20th, 2021. Data processing involved peak picking, run alignment and normalisation, with singly charged spectra excluded from the processing pipeline. Proteins (filtered results to exclude reverse hits, common contaminants and singly charged ions) containing at least two unique peptides were considered valid. Protein quantification was carried out with Progenesis Q1 for Proteomics v2.0.5556.29015 (Nonlinear Dynamics, UK) following the non-conflicting peptides method.

Identification of differentially expressed proteins. Valid proteins were subjected to two-way ANOVA in an R script to compare their expression between the two genotypes in response to each treatment. P-values were adjusted for multiple testing using the Benjamini-Hochberg method. Only proteins that showed a q-value < 0.05 were retained. Log2 fold-change of significant proteins was computed for each genotype in response to each treatment to identify differentially expressed proteins within genotypes under each treatment. Proteins that showed a log2 fold-change > 1 and a q-value < 0.05 were considered differentially expressed. Subsequently, a fold-change ratio was calculated for each protein under each treatment by dividing the Log2 fold-change of Alisto by the Log2 fold-change of ICSB to identify differentially expressed proteins that behave differently between the two genotypes under the respective treatment. Proteins with a log2 fold-change ratio > 1 were considered differentially expressed between the two genotypes. Furthermore, differentially expressed proteins under the combined stress were cross-compared to the differentially expressed proteins in response to the single stresses, and a list of differentially expressed proteins that were unique for the combined treatment was obtained. The lists resulting from the above-mentioned analyses contained differentially expressed proteins between the two genotypes (log2 fold-change ≤ 1) in the other, or an opposite trend observed in the other genotype, or differential levels of higher expression or lower expression between the two genotypes. In other words, a protein may show higher expression (or lower expression) in both genotypes, but the degree of increased expression (or decreased expression) in one genotype should differ by at least two-fold greater than in the other genotype. Furthermore, differentially expressed proteins were further sorted using the conditional formatting and sort features in Excel to identify those that were uniquely upregulated or downregulated for each genotype under each condition. R plots were produced to visualize the numbers of differentially regulated proteins between the two sorghum genotypes in leaves and roots under the three stress conditions as well as the differentially regulated proteins that are unique for combined water deficit and heat stress. Heat maps were generated in R to illustrate the expression levels of differentially regulated proteins between the two genotypes in response to the three stress conditions.

Functional annotation of differentially expressed proteins. Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analyses were conducted using OmicsBox software version 3.0.30¹³, as described by Ali *et al.*¹². The FASTA sequences of differentially expressed proteins were retrieved from the UniProtKB database on June 23rd, 2023. Sequences were searched using DIAMOND BLASTP to determine similarity matches in the database. The BLAST search was performed using the default parameters with a maximum of 20 hits, at an expectation value of 1.0×10^{-3} , with 33 as the high-scoring segment pair (HSP) length cutoff and 0 as the HSP-hit coverage, with application of a low complexity filter. Sequences that received the best BLAST hits were mapped and annotated using default settings (annotation cutoff 55, GO weight 5, e-value 1.0×10^{-6} , HSP-hit coverage cutoff 0 and hit filter 500). Proteins were annotated according to gene ontology (GO) terms, namely biological process (BP), molecular function (MF) and cellular component (CC). Additional functional analyses were performed using the EggNOG-mapper program implemented in the OmicsBox software and the annotation outputs were incorporated to the existing results to enhance the functional annotation of differentially expressed proteins as illustrated in Fig. 1. For instance, Fig. 2, which illustrates the functional analyses of uniquely upregulated proteins in the leaves of Alisto under water deficit stress, shows enhanced GO and

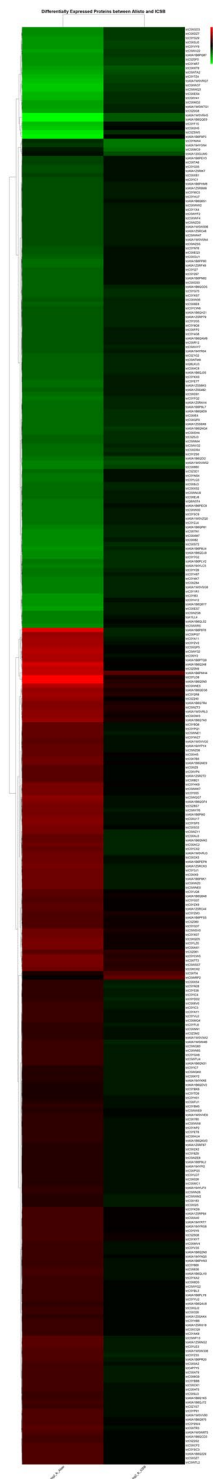


Fig. 13 Presentation of differentially expressed proteins between two contrasting sorghum genotypes under heat stress as observed in the roots. Each column represents the log₂-fold change of differentially expressed proteins in each genotype. Expression differences are illustrated in different colors. Red color indicates upregulation, black color indicates no change in expression and green color indicates downregulation.

enzyme code (EC) annotations after merging the EggNOG-derived annotations with the initial data. Annotated sequences were assigned to KEGG pathways via both Enzyme Commission (EC) number and KEGG Ortholog (KO) identifiers. For KEGG configuration, filtering options included metabolism, genetic information processing, environmental information processing and cellular processes.

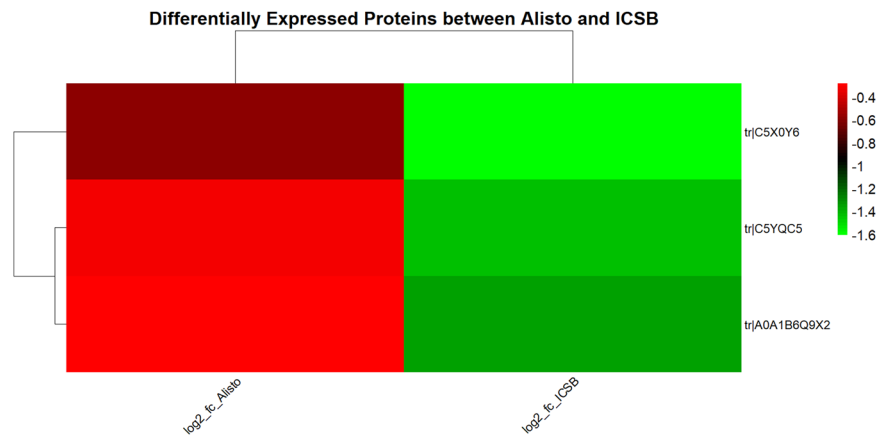


Fig. 14 Presentation of differentially expressed proteins between two contrasting sorghum genotypes under combined water deficit and heat stress as observed in the roots. Each column represents the log2-fold change of differentially expressed proteins in each genotype. Expression differences are illustrated in different colors. Red color indicates upregulation, black color indicates no change in expression and green color indicates downregulation.

Supplementary table	List name
Supplementary table 10	L_WD_ Uniquely upregulated proteins in Alisto
Supplementary table 11	L_WD_ Uniquely upregulated proteins in ICSB
Supplementary table 12	L_WD_ Uniquely downregulated proteins in Alisto
Supplementary table 13	L_WD_ Uniquely downregulated proteins in ICSB
Supplementary table 14	L_H_ Uniquely upregulated proteins in Alisto
Supplementary table 15	L_H_ Uniquely upregulated proteins in ICSB
Supplementary table 16	L_H_ Uniquely downregulated proteins in Alisto
Supplementary table 17	L_H_ Uniquely downregulated proteins in ICSB
Supplementary table 18	L_WD + H_ Uniquely upregulated proteins in Alisto
Supplementary table 19	L_WD + H_ Uniquely upregulated proteins in ICSB
Supplementary table 20	L_WD + H_ Uniquely downregulated proteins in Alisto
Supplementary table 21	L_WD + H_ Uniquely downregulated proteins in ICSB
Supplementary table 22	L_ Uniquely upregulated proteins in Alisto occurring only under WD + H
Supplementary table 23	L_ Uniquely upregulated proteins in ICSB occurring only under WD + H
Supplementary table 24	L_ Uniquely downregulated proteins in Alisto occurring only under WD + H
Supplementary table 25	L_ Uniquely downregulated proteins in ICSB occurring only under WD + H
Supplementary table 26	R_WD_ Uniquely upregulated proteins in Alisto
Supplementary table 27	R_WD_ Uniquely downregulated proteins in Alisto
Supplementary table 28	R_WD_ Uniquely downregulated proteins in ICSB
Supplementary table 29	R_H_ Uniquely upregulated proteins in Alisto
Supplementary table 30	R_H_ Uniquely upregulated proteins in ICSB
Supplementary table 31	R_H_ Uniquely downregulated proteins in Alisto
Supplementary table 32	R_H_ Uniquely downregulated proteins in ICSB
Supplementary table 33	R_WD + H_ Uniquely downregulated proteins in ICSB
Supplementary table 34	R_ Uniquely downregulated proteins in ICSB occurring only under WD + H

Table 4. List names of uniquely upregulated/downregulated proteins associated with the supplementary tables that illustrate the Gene Ontology annotation results obtained through OmicsBox.

Data Records

The multi-step gradient parameters for peptide separation are summarized in Table 1. The parameters of mass spectrometry data acquisition are detailed in Table 2. The details of database search for peptide and protein identification with Byonic are shown in Table 3. The raw mass spectrometry files and metadata have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository¹⁴, NCBI PRIDE <http://www.ebi.ac.uk/pride/archive/projects/PXD052255>: PXD052255 (2024)¹⁵. Figures 3 and 4 illustrate the numbers of differentially expressed proteins between the two genotypes for each treatment in leaves and roots, respectively. Figures 5 to 7 show the numbers of uniquely upregulated and downregulated proteins in each genotype under each treatment. Figure 8 illustrates uniquely upregulated and downregulated proteins in each genotype,

Supplementary table	List name
Supplementary table 35	L_WD_Uniquely upregulated proteins in Alisto
Supplementary table 36	L_WD_Uniquely upregulated proteins in ICSB
Supplementary table 37	L_WD_Uniquely downregulated proteins in Alisto
Supplementary table 38	L_WD_Uniquely downregulated proteins in ICSB
Supplementary table 39	L_H_Uniquely upregulated proteins in Alisto
Supplementary table 40	L_H_Uniquely upregulated proteins in ICSB
Supplementary table 41	L_H_Uniquely downregulated proteins in Alisto
Supplementary table 42	L_H_Uniquely downregulated proteins in ICSB
Supplementary table 43	L_WD + H_Uniquely upregulated proteins in Alisto
Supplementary table 44	L_WD + H_Uniquely upregulated proteins in ICSB
Supplementary table 45	L_WD + H_Uniquely downregulated proteins in Alisto
Supplementary table 46	L_WD + H_Uniquely downregulated proteins in ICSB
Supplementary table 47	L_Uniquely upregulated proteins in Alisto occurring only under WD + H
Supplementary table 48	L_Uniquely upregulated proteins in ICSB occurring only under WD + H
Supplementary table 49	L_Uniquely downregulated proteins in Alisto occurring only under WD + H
Supplementary table 50	L_Uniquely downregulated proteins in ICSB occurring only under WD + H
Supplementary table 51	R_WD_Uniquely upregulated proteins in Alisto
Supplementary table 52	R_WD_Uniquely downregulated proteins in Alisto
Supplementary table 53	R_WD_Uniquely downregulated proteins in ICSB
Supplementary table 54	R_H_Uniquely upregulated proteins in Alisto
Supplementary table 55	R_H_Uniquely downregulated proteins in Alisto
Supplementary table 56	R_H_Uniquely downregulated proteins in ICSB
Supplementary table 57	R_WD + H_Uniquely downregulated proteins in ICSB
Supplementary table 58	R_Uniquely downregulated proteins in ICSB occurring only under WD + H

Table 5. List names of uniquely upregulated/downregulated proteins associated with the supplementary tables that illustrate the KEGG pathway results obtained through OmicsBox.

Supplementary figure	List name
Supplementary figure 4	L_WD_Uniquely upregulated proteins in Alisto
Supplementary figure 5	L_WD_Uniquely upregulated proteins in ICSB
Supplementary figure 6	L_WD_Uniquely downregulated proteins in Alisto
Supplementary figure 7	L_WD_Uniquely downregulated proteins in ICSB
Supplementary figure 8	L_H_Uniquely upregulated proteins in Alisto
Supplementary figure 9	L_H_Uniquely upregulated proteins in ICSB
Supplementary figure 10	L_H_Uniquely downregulated proteins in Alisto
Supplementary figure 11	L_H_Uniquely downregulated proteins in ICSB
Supplementary figure 12	L_WD + H_Uniquely upregulated proteins in Alisto
Supplementary figure 13	L_WD + H_Uniquely upregulated proteins in ICSB
Supplementary figure 14	L_WD + H_Uniquely downregulated proteins in Alisto
Supplementary figure 15	L_WD + H_Uniquely downregulated proteins in ICSB
Supplementary figure 16	L_Uniquely upregulated proteins in Alisto only occurring under WD + H
Supplementary figure 17	L_Uniquely upregulated proteins in ICSB only occurring under WD + H
Supplementary figure 18	L_Uniquely downregulated proteins in Alisto only occurring under WD + H
Supplementary figure 19	L_Uniquely downregulated proteins in ICSB only occurring under WD + H
Supplementary figure 20	R_WD_Uniquely upregulated proteins in Alisto
Supplementary figure 21	R_WD_Uniquely downregulated proteins in Alisto
Supplementary figure 22	R_WD_Uniquely downregulated proteins in ICSB
Supplementary figure 23	R_H_Uniquely upregulated proteins in Alisto
Supplementary figure 24	R_H_Uniquely downregulated proteins in Alisto
Supplementary figure 25	R_H_Uniquely downregulated proteins in ICSB
Supplementary figure 26	R_WD + H_Uniquely downregulated proteins in ICSB
Supplementary figure 27	R_Uniquely downregulated proteins in ICSB only occurring under WD + H

Table 6. List names of uniquely upregulated/downregulated proteins associated with the supplementary figures that illustrate the biological process (BP) enrichment results obtained through OmicsBox.

Supplementary figure	List name
Supplementary figure 28	L_WD_ Uniquely upregulated proteins in Alisto
Supplementary figure 29	L_WD_ Uniquely upregulated proteins in ICSB
Supplementary figure 30	L_WD_ Uniquely downregulated proteins in Alisto
Supplementary figure 31	L_WD_ Uniquely downregulated proteins in ICSB
Supplementary figure 32	L_H_ Uniquely upregulated proteins in Alisto
Supplementary figure 33	L_H_ Uniquely upregulated proteins in ICSB
Supplementary figure 34	L_H_ Uniquely downregulated proteins in Alisto
Supplementary figure 35	L_H_ Uniquely downregulated proteins in ICSB
Supplementary figure 36	L_WD + H_ Uniquely upregulated proteins in Alisto
Supplementary figure 37	L_WD + H_ Uniquely upregulated proteins in ICSB
Supplementary figure 38	L_WD + H_ Uniquely downregulated proteins in Alisto
Supplementary figure 39	L_WD + H_ Uniquely downregulated proteins in ICSB
Supplementary figure 40	L_ Uniquely upregulated proteins in Alisto only occurring under WD + H
Supplementary figure 41	L_ Uniquely upregulated proteins in ICSB only occurring under WD + H
Supplementary figure 42	L_ Uniquely downregulated proteins in Alisto only occurring under WD + H
Supplementary figure 43	L_ Uniquely downregulated proteins in ICSB only occurring under WD + H
Supplementary figure 44	R_WD_ Uniquely upregulated proteins in Alisto
Supplementary figure 45	R_WD_ Uniquely downregulated proteins in Alisto
Supplementary figure 46	R_WD_ Uniquely downregulated proteins in ICSB
Supplementary figure 47	R_H_ Uniquely upregulated proteins in Alisto
Supplementary figure 48	R_H_ Uniquely upregulated proteins in ICSB
Supplementary figure 49	R_H_ Uniquely downregulated proteins in Alisto
Supplementary figure 50	R_H_ Uniquely downregulated proteins in ICSB
Supplementary figure 51	R_WD + H_ Uniquely downregulated proteins in ICSB
Supplementary figure 52	R_ Uniquely downregulated proteins in ICSB only occurring under WD + H

Table 7. List names of uniquely upregulated/downregulated proteins associated with the supplementary figures that illustrate the molecular function (MF) enrichment results obtained through OmicsBox.

Supplementary figure	List name
Supplementary figure 53	L_WD_ Uniquely upregulated proteins in Alisto
Supplementary figure 54	L_WD_ Uniquely upregulated proteins in ICSB
Supplementary figure 55	L_WD_ Uniquely downregulated proteins in Alisto
Supplementary figure 56	L_WD_ Uniquely downregulated proteins in ICSB
Supplementary figure 57	L_H_ Uniquely upregulated proteins in Alisto
Supplementary figure 58	L_H_ Uniquely upregulated proteins in ICSB
Supplementary figure 59	L_H_ Uniquely downregulated proteins in Alisto
Supplementary figure 60	L_H_ Uniquely downregulated proteins in ICSB
Supplementary figure 61	L_WD + H_ Uniquely upregulated proteins in Alisto
Supplementary figure 62	L_WD + H_ Uniquely upregulated proteins in ICSB
Supplementary figure 63	L_WD + H_ Uniquely downregulated proteins in Alisto
Supplementary figure 64	L_WD + H_ Uniquely downregulated proteins in ICSB
Supplementary figure 65	L_ Uniquely upregulated proteins in Alisto only occurring under WD + H
Supplementary figure 66	L_ Uniquely upregulated proteins in ICSB only occurring under WD + H
Supplementary figure 67	L_ Uniquely downregulated proteins in Alisto only occurring under WD + H
Supplementary figure 68	L_ Uniquely downregulated proteins in ICSB only occurring under WD + H
Supplementary figure 69	R_WD_ Uniquely upregulated proteins in Alisto
Supplementary figure 70	R_WD_ Uniquely downregulated proteins in Alisto
Supplementary figure 71	R_H_ Uniquely upregulated proteins in Alisto
Supplementary figure 72	R_H_ Uniquely upregulated proteins in ICSB
Supplementary figure 73	R_H_ Uniquely downregulated proteins in Alisto
Supplementary figure 74	R_H_ Uniquely downregulated proteins in ICSB
Supplementary figure 75	R_WD + H_ Uniquely downregulated proteins in ICSB
Supplementary figure 76	R_ Uniquely downregulated proteins in ICSB only occurring under WD + H

Table 8. List names of uniquely upregulated/downregulated proteins associated with the supplementary figures that illustrate the cellular component (CC) enrichment results obtained through OmicsBox.

Supplementary figure	List name
Supplementary figure 77	L_WD_Uniquely upregulated proteins in Alisto
Supplementary figure 78	L_WD_Uniquely upregulated proteins in ICSB
Supplementary figure 79	L_WD_Uniquely downregulated proteins in Alisto
Supplementary figure 80	L_WD_Uniquely downregulated proteins in ICSB
Supplementary figure 81	L_H_Uniquely upregulated proteins in Alisto
Supplementary figure 82	L_H_Uniquely upregulated proteins in ICSB
Supplementary figure 83	L_H_Uniquely downregulated proteins in Alisto
Supplementary figure 84	L_H_Uniquely downregulated proteins in ICSB
Supplementary figure 85	L_WD + H_Uniquely upregulated proteins in Alisto
Supplementary figure 86	L_WD + H_Uniquely upregulated proteins in ICSB
Supplementary figure 87	L_WD + H_Uniquely downregulated proteins in Alisto
Supplementary figure 88	L_WD + H_Uniquely downregulated proteins in ICSB
Supplementary figure 89	L_Uniquely upregulated proteins in Alisto occurring only under WD + H
Supplementary figure 90	L_Uniquely upregulated proteins in ICSB occurring only under WD + H
Supplementary figure 91	L_Uniquely downregulated proteins in Alisto occurring only under WD + H
Supplementary figure 92	L_Uniquely downregulated proteins in ICSB occurring only under WD + H
Supplementary figure 93	R_WD_Uniquely upregulated proteins in Alisto
Supplementary figure 94	R_WD_Uniquely downregulated proteins in Alisto
Supplementary figure 95	R_WD_Uniquely downregulated proteins in ICSB
Supplementary figure 96	R_H_Uniquely upregulated proteins in Alisto
Supplementary figure 97	R_H_Uniquely downregulated proteins in Alisto
Supplementary figure 98	R_H_Uniquely downregulated proteins in ICSB
Supplementary figure 99	R_WD + H_Uniquely downregulated proteins in ICSB
Supplementary figure 100	R_Uniquely downregulated proteins in ICSB occurring only under WD + H

Table 9. List names of uniquely upregulated/downregulated proteins associated with the supplementary figures that illustrate the KEGG pathways and the number of sequences linked to each pathway (obtained through OmicsBox).

which occur only under combined water deficit and heat stress. Notably, Fig. 3 shows a total of 175 differentially expressed proteins between the leaves of the two genotypes under combined water deficit and heat stress. However, in Fig. 7 (a and b), the total number of differentially expressed proteins in the leaves is 176. This is because one protein identified as uniquely upregulated in the leaves of the Alisto genotype is uniquely downregulated in ICSB (Supplementary Table 4). Therefore, this protein is counted twice in the diagram, leading to a total count of 176. Figures 9 to 14 present heat maps showing differentially expressed proteins between the two genotypes in both leaves and roots under water deficit, heat, and combined water deficit and heat stress. The processed data, including the list of all valid proteins with their relative quantification, lists of differentially expressed proteins between the two sorghum genotypes for each tissue and each treatment as well as GO and KEGG result files from OmicsBox are openly available as supplementary data at FigShare¹⁶ under <https://doi.org/10.6084/m9.figshare.25962400.v3>. Supplementary Table 1 shows all the valid proteins with their relative abundance in leaf and root tissues for each genotype and treatment. The table is provided with the accessions of proteins, peptide count, number of unique peptides, confidence score and description of proteins. In the table, the following abbreviations are used to indicate different experimental conditions and tissue types: WW (well-watered/normal condition), WD (water deficit), H (heat), WD + H (combined water deficit and heat), L (leaf tissue), and R (root tissue). Supplementary Tables 2 to 9 represent the lists of differentially expressed proteins between the two genotypes for L_WD; L_H; L_WD + H; R_WD; R_H; R_WD + H; unique proteins for L_WD + H and unique proteins for R_WD + H, respectively. In these tables, the title of each worksheet indicates the respective step in the differential expression analysis. Notably, as shown in Fig. 5c, our analyses did not identify any uniquely upregulated proteins in ICSB roots under water deficit. In addition, the root analyses under combined water deficit and heat stress only resulted in uniquely downregulated proteins in ICSB (Figs. 4 and 8). Supplementary Tables 10 to 34 are OmicsBox annotation tables, providing detailed annotation results of the lists of uniquely upregulated/downregulated proteins for each genotype under each treatment as detailed in Table 4. Within these files, column C presents the accessions of proteins, column D provides descriptions of the proteins, columns J and K indicate the IDs and names of the GO terms associated with each protein, columns L and M reflect the enzyme code and name of each sequence, column N denotes the InterPro IDs, and columns O and P represent the IDs and names of the InterPro GO terms. Additional description of the data included in these tables can be found at <https://manual.omicsbox.biobam.com/user-manual/omicsbox-modules/module-functional-analysis/gene-ontology-annotation/#GeneOntologyAnnotation-ResultTable>. Supplementary Tables 35 to 58 are OmicsBox pathway tables, presenting the KEGG pathways associated with the exclusively upregulated/downregulated proteins in the two genotypes under each treatment as detailed in Table 5. Notably, the uniquely upregulated proteins observed in the roots of ICSB under heat stress were not linked to any KEGG pathways. Within these files, column A denotes the pathway name, column B indicates

the Pathway ID, column C reflects the number of sequences involved in each pathway, column D indicates the number of sequences linked to KEGG pathways through enzyme code (EC), column E indicates the number of sequences linked to KEGG pathways via KEGG Ortholog (KO), column F indicates the accession numbers of proteins, column G denotes the EC number of each sequence, and column H represents the KO number for each sequence. In addition, these files include graphs visualizing the KEGG pathways and the number of sequences linked to each pathway. Furthermore, Supplementary Figure 1 demonstrates the relative water content (RWC) in the leaves of the two sorghum genotypes under each growth condition. Supplementary Figure 2 shows the chlorophyll content in the leaves of both genotypes under each condition. Supplementary Figure 3 displays the shoot length (a) and stem diameter (b) of the plants under each condition. Supplementary Figures 4 to 27 illustrate the biological processes (BP) associated with the lists of differentially expressed proteins outlined in Table 6. Notably, no biological processes were enriched for the uniquely upregulated proteins observed in the roots of ICSB under heat stress. Similarly, Supplementary Figures 28 to 52 present the molecular functions (MF) associated with the lists of differentially expressed proteins described in Table 7. Supplementary figures 53 to 76 describe the cellular components (CC) associated with the lists of differentially accumulating proteins outlined in Table 8. There were no cellular components enriched for the uniquely downregulated proteins observed in the roots of ICSB under water deficit stress. Additionally, supplementary figures 77 to 100 represent the KEGG pathways and the number of sequences linked to each pathway for the lists of differentially expressed proteins as detailed in Table 9.

Technical Validation

We performed an SDS-PAGE to confirm that the protein samples were not degraded before the LC-MS/MS analyses. Pooled samples were run at intervals between samples as quality control measures to correct for drift in retention time across samples and thus ensure the performance of LC-MS/MS data acquisition.

Usage Notes

The annotation tables of differentially expressed proteins can be visualized in Excel to examine the GO terms associated with the proteins that showed differential expressed between the two genotypes for each tissue and treatment. Additionally, these tables allow for the identification of GO terms associated with proteins that were solely differentially expressed between the two genotypes under the combined stress, both in leaves and roots, elucidating the unique responses to the combined stress. Similarly, pathway tables can be viewed in Excel to explore KEGG pathways associated with the differentially expressed proteins between the two genotypes under each treatment. Proteins that were upregulated or downregulated in one genotype but not the other represent the distinct responses exhibited by each genotype for each treatment. For example, the annotation results of uniquely upregulated proteins in Alisto elucidate the biological processes and pathways that were uniquely activated in Alisto when compared to ICSB. This dataset provides valuable information for the investigation of the molecular mechanisms of sorghum responses to the single and combined effects of drought and heat stress. Researchers can use the data to explore stress-specific protein expression patterns, identify candidate proteins for functional validation, and conduct comparative analyses across genotypes or plant species. These applications could facilitate the development of sorghum varieties and possibly other species with superior tolerance to these environmental stresses.

Code availability

No custom code was used to generate the data described in this study.

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Author contributions

Conceptualization, N.L.; methodology, A.E.E.A., N.L., L.G., R.E.S., S.C.P. and D.L.T.; software, A.E.E.A.; validation, N.L. and A.E.E.A.; formal analysis, A.E.E.A.; investigation, A.E.E.A. and N.L.; resources, N.L. and A.E.E.A.; data curation, A.E.E.A. and N.L.; writing—original draft preparation, A.E.E.A.; writing—review and editing, A.E.E.A., N.L., R.E.S., L.G., S.C.P. and D.L.T.; visualization, A.E.E.A.; supervision, N.L. and R.E.S.; project administration, N.L.; funding acquisition, N.L. All authors have read and agreed to the published version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to N.L.

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